Maintenance

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BioFocus Preventive Maintenance

For optimum BioFocus system performance, follow the maintenance schedule listed below.

Weekly Maintenance

1. Check the coolant level and add coolant if required. Add 2 ml of methanol, isopropyl alcohol, or ethanol to the coolant reservoir if water is being used.

2. Clean the cartridge compartment. See "Cleaning Cartridge Compartment" on page 6-4.

3. Clean the top hole in the pressure chamber and apply a new, very thin vacuum grease coating. See "Cleaning the Pressure Chamber" on page 6-4.

4. Run defrag.exe or other Windows compatible disk defragmenter.

5. Clean the pressure chamber seals. See "Cleaning the Pressure Chamber" on page 6-4.

6. Empty the condensation tray. Refer to "Emptying the Condensation Tray" on page 6-8.

Semiannual Maintenance

1. Clean the air filter underneath the instrument with a vacuum cleaner. Tilt the instrument back to provide access to the large hole in the center. Be careful not to tip the instrument over.

2. Clean vial holders in carousel.

3. Clean Pressure Chamber/Inlet Electrode interior.

4. Replace the Coolant and Coolant Filter in Reservoir. See "Changing Coolant and the Coolant Reservoir Filter" on page 6-5.

5. Remove the two air filters in the rear of the Sample Area and clean them by blowing air through them.

LIF^2 Detector Preventive Maintenance

To provide peak LIF^2 Detector performance, follow the preventive maintenance schedule outlined below:

Weekly Maintenance

1. Run the Light Level diagnostics from the main BioFocus screen.
2. Inspect the air filter located on the right side of the cabinet. If there is a dust accumulation on the filter's outer surface, remove the dust with a vacuum cleaner.

Semiannual Maintenance

1. Clean the cabinet exterior with a mild commercial detergent.
2. Run the LIF performance test.

BioFocus System Maintenance

Use the information in the following paragraphs to perform semiannual preventive maintenance and run diagnostics.

Cleaning Cartridge Compartment

The cartridge compartment and all components should always be kept as clean as possible to avoid arcing and current leakage during high voltage operation. Surfaces contaminated with buffer residue and salt crystals become conductive and decrease the uppermost voltage level at which the instrument can operate. Current leakage and arcing due to contamination will cause inaccuracies and intermittent problems. To avoid these problems, the cartridge area must be clean.

Remove the front cover of the cartridge holder by loosening the thumbscrews securing it. Clean the area where the cartridge is placed with a cotton swab and deionized water, then wipe the area dry with clean swabs. If another solvent is desired, use isopropyl alcohol, methanol, or ethanol. Use of other organic compounds can cause drift over long periods of time.

Cleaning the Pressure Chamber

It is advisable to remove the Pressure Chamber/Inlet Electrode occasionally for thorough cleaning. Swab out the inside and clean the top with methanol, isopropyl alcohol, or ethanol. Clean the lip seals in the lower portion and inspect them for damage. Apply a very thin vacuum grease coating around the inside diameter of the small hole in top of the chamber. This ensures an effective electrical barrier.

At the lower rim of the pressure chamber there are two seals that will wear over time, necessitating eventual pressure chamber replacement for efficient purging and sample pressure injection. These seals should be cleaned with a cotton swab and water every month to ensure proper operation.

Cleaning the Capillary Cartridge

The Shutdown Template in the Methods section can be used to both clean and dry the capillary and turn off the detector lamp at the end of an automation sequence, extending both the lifetime of the capillary and the detector lamp. This is particularly critical at high pH or when working with large molecules. Be sure that your Configuration's inlet carousel has stations for wash solution, deionized water, and gas (empty).

Cleaning Carousels and Replacing O-Rings

Periodically, the carousels should be emptied and cleaned to prevent accumulation of contaminants that may interfere with sample data collection and high voltage isolation. The carousels may be immersed in a bath of warm water and mild detergent. Then all carousel parts should be rinsed thoroughly with deionized water and allowed to dry prior to installation in the BioFocus carousel compartment. Check that none of the O-rings have slipped off the caps during cleaning. Wipe the upper ends of the frequently-used vial holders to reduce the possibility of arcing and help keep the pressure chamber/inlet electrode clean.

Changing Coolant and the Coolant Reservoir Filter

During routine instrument use, the coolant reservoir level should be regularly checked and coolant replenished as needed. The coolant level should always be between the two grooves on the wall of the reservoir. Weekly addition of 1 to 2 ml of methanol (or another noncorrosive, nonconductive antimicrobial agent) to the coolant reservoir will help prevent bacterial growth. Figure 6-1 shows coolant circulation path.

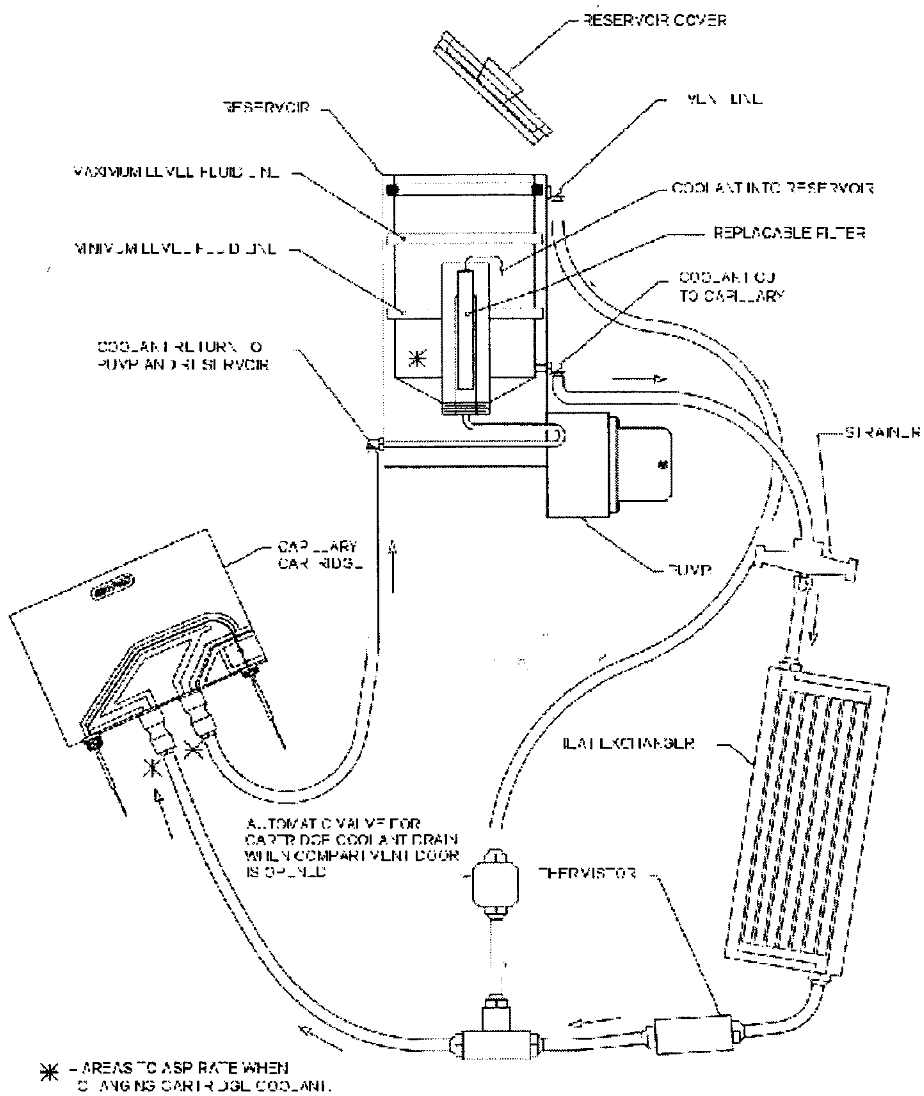


Figure 6-1 Coolant Circulation Path

For operation at voltages below 20 kV, deionized water is the recommended coolant. For most operations between 20-25 kV with the User-Assembled Cartridge, water can be used as coolant. For operation at 25 kV and above with the User-Assembled Cartridge, water should be replaced with Fluorinert FC 77, Bio-Rad catalog #148-6050. A small bottle of this has been provided with your system.

A separate filter should be dedicated for use with Fluorinert because it is difficult to remove water completely from the filter matrix (extra 60 µm filters are supplied with the system).

Change coolant by using the following procedure:

1. Unscrew the filter holder from the coolant reservoir using a wide blade screwdriver. Remove the filter element from the bottom of the holder.

2. Withdraw as much of the liquid from the bottom of the reservoir as possible using the Coolant Reservoir Purge Device. Use a cotton swab to wipe any residual moisture from the reservoir.

3. Make sure that a capillary cartridge is in the cartridge holder and the door is closed. Insert the syringe adapter into the reservoir and screw it into place. Withdraw the remaining coolant in the lines by suction (pulling back the syringe plunger. See Figure 6-2.

NOTE: Do not apply pressure—coolant lines inside the system could become disconnected.

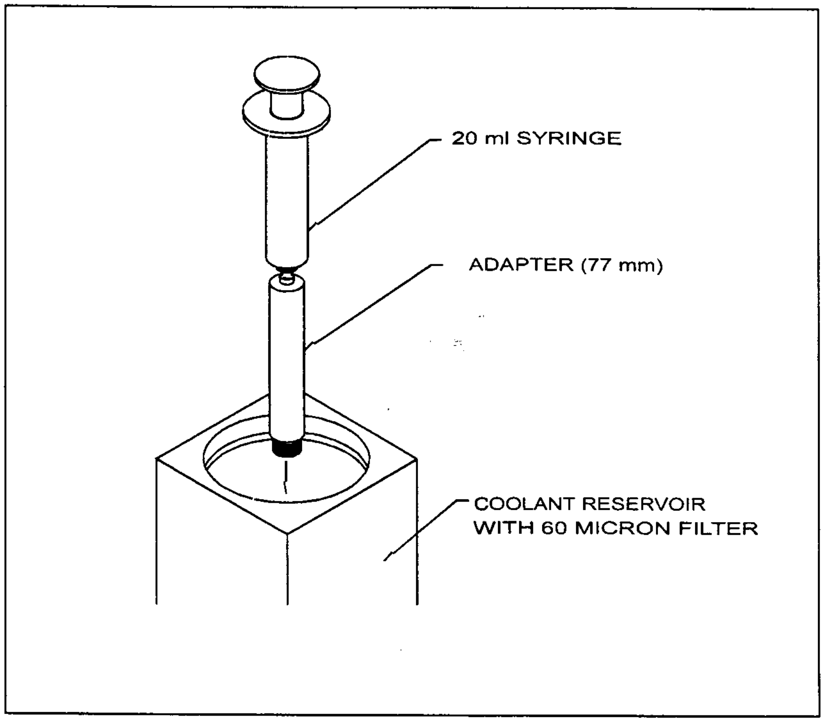


Figure 6-2 Syringe and Reservoir

If switching to Fluorinert from water, fill the reservoir with methanol and start the pump from Temperature Diagnostics. Run for 2 minutes then aspirate most of the methanol from the system. Now add Fluorinert. Because Fluorinert is heavier than water, any residual water in the system floats to the surface and sides of the reservoir; it should be removed with a cotton swab or syringe.

NOTE: Having residual Fluorinert in the system may cause the pump to stop unexpectedly.

If switching to water from Fluorinert, fill the reservoir with water, then start the pump from the Temperature Diagnostics screen. Place the tip of the syringe at the base of the reservoir and withdraw residual Fluorinert from the bottom until the reservoir is almost empty. Repeat this process twice more; at this point the liquid should be completely clear with no immiscible Fluorinert observable in the water coolant.

Install a new filter in the filter holder (hole should be up, visible in slot) and screw it into place.

Emptying the Condensation Tray

There is a tray located underneath the BioFocus instrument. This tray catches water that condenses from the cartridge cooling system. You should check the condensation tray every week, especially if the ambient humidity is high. Reach under the front of the instrument and pull the tray out partially. If there is water in the tray, hold a container underneath the right front edge of the tray and push the tray's right front corner down to drain the water.

Checking System Light Levels

The lamp assemblies supplied with the BioFocus systems are prealigned; no adjustments are needed. The D_2 lamp is used for operations in the 190-365 nm range and is warranted for 500 hours. Each D_2 lamp assembly is equipped with a chronometer indicating the total hours of operation. The chronometer is read by noting the position of the gap in the mercury tube against the graduated background. The tungsten lamp is used for operations in the 366-800 nm range and is warranted for 90 days. The tungsten lamp is optional in the BioFocus 2000.

The BioFocus optical system employs two separate photodiodes. The light from the monochromator is split into two beams. One is directed to the reference photodiode and the other beam passes through the capillary and strikes the sample photodiode. Light levels may be monitored for both; they are displayed as Reference V to F, Sample V to F, and Ratio (%) in the Light Levels Diagnostics screen.

Checking the Deuterium or Tungsten Lamp

Power up the system and start the BioFocus software. The carousel and cartridge compartment doors must be closed. Wait at least 30 minutes if you have just turned on the system. Go to **Diagnostics**, then **Light Levels** under the **Commands** menu. To check the D_2 lamp, enter 254 nm in the Wavelength Test box, and then click to the right of this box. This turns the D_2 lamp on. To check the tungsten lamp, enter a number greater than 366 nm in the **Wavelength Test** box, then click to the right of this box. This will turn the tungsten lamp on.

NOTE: Light levels in the BioFocus 3000 and the BioFocus 2000 are different because the sampling time is shorter on the BioFocus 3000 in scanning mode, which is used in the diagnostic test.

If after 30 seconds the Reference V to F reading is equal to or greater than 100 on the BioFocus 3000 (or greater than 200 on the BioFocus 2000), the D_2 lamp is operating effectively. If the displayed value is less than 100 on the BioFocus 3000 (or less than 200 on the BioFocus 2000), the lamp should be replaced.

Replacing Detector Lamps

If you need to replace a D_2 lamp, order Bio-Rad catalog number 167-0905. For a tungsten lamp, order catalog number 167-0906.

Make sure that the power switch on the back of the instrument is off before opening the lamp compartment door. To remove the lamp cover, turn the thumbscrew until the spring pushes the screw back, then remove the cover.

WARNING! UV light can damage eyes and skin. Always disconnect the power cord before working in the vicinity of the lamp. The D_2 lamp becomes very hot. Care must be taken while handling it to prevent burns. Always allow the lamp to cool before removing it.

The lamp compartment is illustrated in Figure 6-3, showing the locations of lamp assemblies, mounting screws, plugs, and alignment pins.

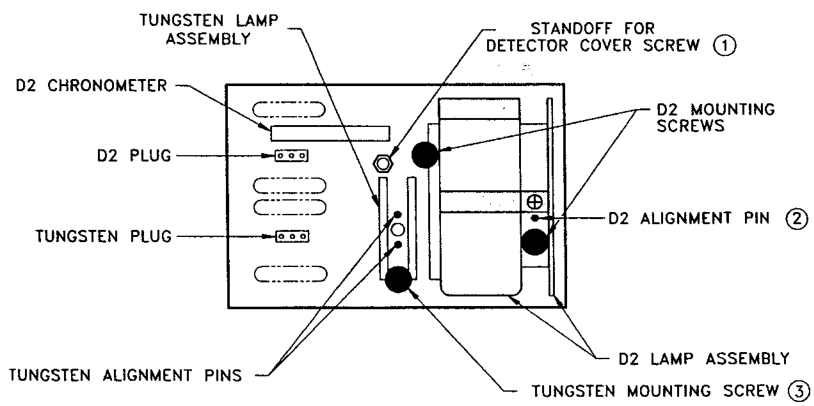


Figure 6-3 Lamp Compartment

Changing The Deuterium Lamp

To change the D_2 lamp, disconnect the upper lamp plug (black leads) by gently pulling it straight back toward you. ***Do not twist*** the connector while pulling. Unscrew the two black thumbscrews holding the lamp assembly in place (one on the lower right and one on the upper left side of the lamp) and pull the lamp assembly straight back toward you.

NOTE: Do not touch the glass bulb of the D2 lamp. Any contamination can decrease the life of the lamp

To install a new lamp, slide the lamp assembly onto the alignment pin (the assembly has a predrilled hole to accommodate the pin).

If your fingers touched the glass, carefully wipe off any fingerprints with alcohol. Do not loosen the screw or metal bracket holding the lamp to the assembly, and do not attempt to rotate or move the lamp up or down in the assembly (doing so will degrade performance).

Use the thumbscrews to attach the lamp assembly back to the detector. Connect the lamp plug back to the upper terminal of the lamp compartment.

Changing The Tungsten Lamp

To change the tungsten lamp, disconnect the lower lamp plug (white leads) by gently pulling it straight back toward you. *Do not twist* the connector while pulling. Unscrew the lower black thumbscrew holding the lamp assembly in place and pull the lamp assembly straight back towards you.

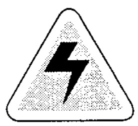
To install a new lamp, slide the lamp assembly back onto the two alignment pins.

Connect the lamp plug back to the lower terminal in the lamp compartment.

Replace the lamp cover before turning BioFocus system power on. Lift the cover slightly, align the screw, push in the spring, then tighten the screw.

Replacing Fuses

BioFocus 2000/2000TC



WARNING! To avoid electrical shock, unplug the instrument before changing fuses.

NOTE: Fuses may have to be changed during reconfiguration of the power entry module.

There are two fuse locations on the BioFocus 2000 and 2000TC; both are accessible on the upper rear of the instrument. Refer to Figure 6-4. One fuse protects the detector portion of the instrument and the fuses located in the power entry module protect the instrument in total. The proper fuses for the operating voltage are indicated on the rear of the instrument.

Remove the cover of the power entry module by pushing the catch to the right. The cover should then pop out slightly for easy removal. The fuses are retained in the cover. Replace as required.

Replace the power entry module cover by pushing it straight in until the catch clicks.

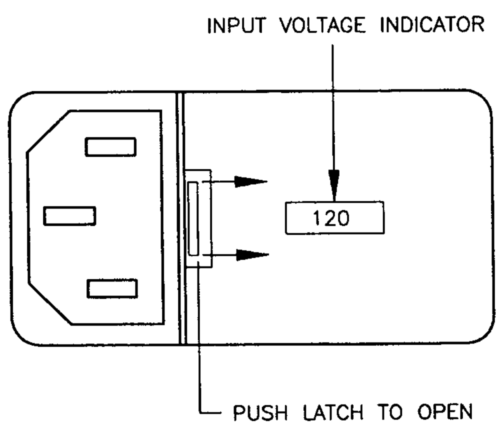


Figure 6-4 BioFocus 2000 Power Entry Module

BioFocus 3000/3000TC

WARNING! To avoid electrical shock, unplug the instrument before changing fuses.

To replace fuses, remove the power entry module cover using a small flat-bladed screw driver, prying outward in the slot provided. Remove the fuses using a fuse puller. (A paper clip bent with a hook on the end can also be used for pulling fuses). See Figure 6-5.

The BioFocus 3000 and 3000TC have two fuse configurations each. One is for 100 or 120 VAC operation and one is for 220 or 240 VAC operation. The lower range uses one of the 1/4 in. diameter (large) fuses in the upper fuse position and a bus bar that is retained in place of a smaller 5 mm fuse in the lower fuse position. The high voltage configuration uses two of the 5 mm diameter (small) fuses. The fuses are located in the power entry module which is located on the upper rear panel of the instrument.

Install the proper fuses by pushing into place with your fingers. The rear fuses and bus bar may need to be pushed in fully with the eraser end of a pencil. Take care not to push the larger fuses too far rearward because this may damage the fuse retainer. The larger fuse should sit just behind the cover.

NOTE: Check that the white plastic indicator shows through the cover in the correct position

Replace the cover by pushing it straight in until it snaps into place.

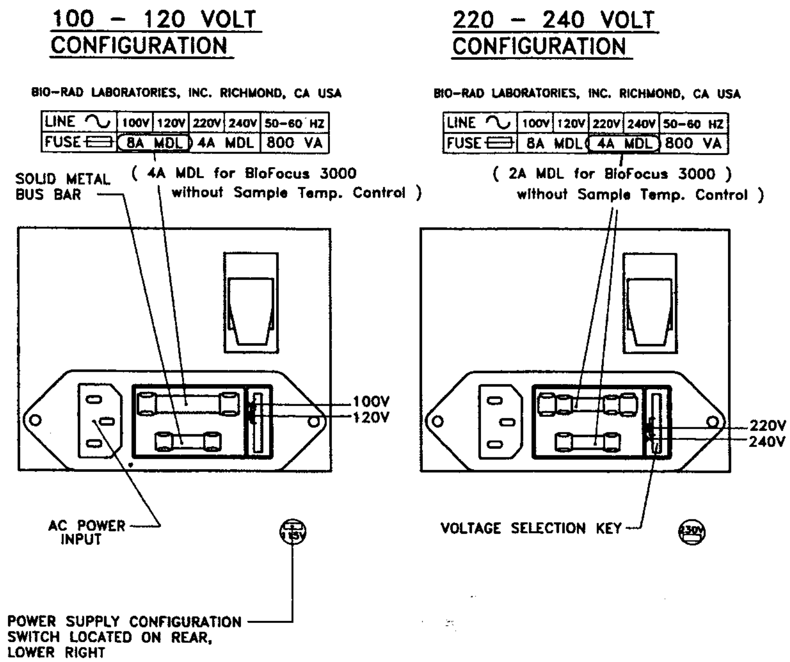
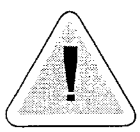


Figure 6-5 BioFocus 3000 Power Entry Module



CAUTION! Failure to select the proper fuse for your instrument's operating voltage may damage the instrument.

Troubleshooting

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Introduction

This section contains troubleshooting information. It includes tables that list various systems and probable causes.

Diagnostics that are part of the operating software lead you through system checks once you have isolated problems.

Troubleshooting Guide

Absorbance Mode Troubleshooting.

Table 7-1 lists system problems, possible causes, and suggested remedies.

Table 7-1 Symptoms and Probable Causes

Symptom	Possible Cause	Remedy
No peaks during run	Air bubble in capillary	Refill vials.
	Run time too short	Increase run time.
	Damaged capillary	Replace capillary.
	Defective D2 lamp	Run light level diagnostics and replace lamp if required.
	Buffer concentration too high	Dilute electrolyte or use lower voltage.
	Sample vial in wrong position in carousel	Check sample vial position and move if required.
	Wrong voltage polarity	Check method and voltage settings.

Symptom	Possible Cause	Remedy
	Sample not injected	Low gas pressure or high voltage power supply defective. Run pressure and electrical diagnostics.
	Low vial levels	Check vial levels and replenish if required.
	High voltage current leakage	Clean cartridge area to remove contamination.
	Blocked capillary	Purge or replace capillary.
Poor sensitivity	Low sample concentration	Increase sample injection time.
	Background UV absorbance	Use a different buffer.
Poor peak area accuracy	Sample contamination	Rinse capillary tip with a dip cycle.
	Sample evaporation	Keep sample vials covered until ready for a run.
	Small peak size	Use higher sample concentration or longer sample injection time.
Poor peak shape or height	High sample concentration	Dilute sample.
	Joule heating	Decrease buffer ionic strength and/or decrease run voltage.
		Use capillary with smaller ID.

Symptom	Possible Cause	Remedy
	Incorrect electrolyte concentration	Optimize electolyte concentration when developing the method to be used.
	Incorrect voltage or current	Optimize the voltage or current when developing the method.
	Long injection time	Reduce injection time if possible
Migration time variation	Buffer depletion	Use larger vial volume.
	Different standard and sample composition	Match sample and standard ionic strengths and solvent contents.
	Electroosmotic flow variation	Use proper capillary conditioning and rinsing. Dedicate one cartridge to each method.
	Incomplete equilibration	Use one or two pre-injection cycles to allow capillary wall and temperature to stabilize.
	Incorrect cartridge temperature	Check cartridge temperature specified in the method.
	Unequal vial levels	Sample and/or buffer siphoning. Check levels and equalize prior to a run.

Symptom	Possible Cause	Remedy
Nothing happens when all power switches are turned on.	Power cord not connected properly	Check power cord connections.
	Fuse Blown	Check and replace fuses if needed.
Carousel jams.	Carousel improperly seated	Push Reset on rear panel; remove and reinsert carousel. Be sure to push carousel near top when inserting.
Cooling rate too slow, or not cooling enough	Bad coolant circulation	Check for air bubbles and verify flow in coolant reservoir. Replace reservoir filter.
	Obstructed fan under unit	Make sure there is nothing obstructing fan inlet underneath the instrument.
Mouse frozen.	Windows software fault	Reboot computer and restart Windows. Start BioFocus software again. If condition persists, contact a service representative.
Automation sequence interrupted.	A compartment door is open	Check to make sure both doors are closed properly

Symptom	Possible Cause	Remedy
Low sample light level.	Bad cartridge	See light level diagnostics
	Wavelength setting too low for the type of buffer in the capillary	Use a higher wavelength setting
Low reference light level.	Lamp is past useful life	Replace lamp
Poor peak migration time reproducibili ty.	Cartridge coolant path partially blocked	Check temperature trace in post-run files. If staircase oscillation of temperature is seen, replace cartridge
	Bubbles in the capillary	Degas buffers by centrifuging and purge capillary.
	Problem with power supply	Check voltage and current trace on bff file
	Electrical leakage	Clean cartridge area

Symptom	Possible Cause	Remedy
"Pressure Out of Range" message at power up or during an automation sequence	Empty gas tank.	Replace gas tank.
	External tubing and fittings not properly connected	Check connections on gas tank and rear panel of the BioFocus 3000 system
	Internal pneumatics module	Call your Bio-Rad representative
	Broken capillary cartridge	Replace cartridge
	Worn seals in pressure chamber	Replace pressure chamber
	Internal pneumatics module failure or tubing blocked	Call your Bio-Rad representative
	Line from pressure regulator too long	Use shorter line from gas tank

Symptom	Possible Cause	Remedy
Unstable or no current during injection or run, but voltage reading is OK	Insufficient purge time	Increase purge time
	Blocked capillary	Purge capillary, replace cartridge if necessary
	Broken capillary inside cartridge	Replace cartridge
	One of the vials is empty	Check vials
	Broken or loose electrode	Check both electrodes and replace an electrode if necessary
	Joule heating	Decrease buffer ionic strength and/or decrease run voltage Use capillary with smaller ID
	Sample in aqueous solution (water plug enters run buffer during pressure injection)	Prepare sample in dilute run buffer. Normally a 1:10 dilution of run buffer is used
Voltage below set point during inject or run in constant voltage mode	Current has reached the set limit.	Reset the current limit or check buffers for proper molarity. Run electrical diagnostics

Symptom	Possible Cause	Remedy
Noisy baseline	Arcing or current leakage across dirty cartridge area	Test baseline with high voltage power off. If noise stops, clean cartridge area
	Capillary cartridge with internal cracks leaking current	Replace cartridge
	Loose outlet electrode	Replace outlet electrode
	Unstable or deteriorated lamp	Check reference light level—replace lamp if necessary
	Capillary cartridge with contaminated optics	Replace cartridge
	Too much Joule heating in the capillary, caused by high current	Reduce voltage or current setting, reduce buffer concentration, or decrease programmed cartridge temperature

Symptom	Possible Cause	Remedy
No peaks and no detector response.	Wrong polarity, sample not migrating past detector	Reverse polarity
	Bad capillary cartridge	Check light levels and replace cartridge if necessary
Peaks smaller than expected.	AU range setting too high	Reset AU Range in Method Detector setup
	Option switches on rear panel set improperly	Set proper signal output value on rear panel
	Leakage in low pressure pneumatics	Check low pressure in Diagnostics.
	Wrong wavelength	Reset wavelength
Broad or asymmetrica l peaks, decreasing peak heights	Contamination of sample vial from buffer residue on the electrode and the capillary inlet	Use a rinse step after the run buffer (water purge for 0 seconds) and frequent buffer changes
	Breakdown of capillary coating	Replace capillary cartridge

Symptom	Possible Cause	Remedy
Programmed Cartridge Temp not attained, bubbles in coolant reservoir, and "low flow" error message	Coolant path in cartridge is plugged or leaking.	Check with purge device and/or replace cartridge and test temperature again.
	Plugged coolant filter or filter upside down	Replace filter with hole up in coolant reservoir
	Leaks or blockages elsewhere in coolant path	Should a problem persist after replacing the cartridge and filter, call your Bio-Rad representative
	Bacterial growth has clogged system	Replace filter and coolant

LIF Mode Troubleshooting

Table 7-2 lists LIF mode symptoms and probable causes.

Table 7-2 LIF Mode Troubleshooting

Symptom	Possible Cause	Remedy
Power-on LED not lighted	AC power not connected	Check AC power connection
	Switching power supply failed	Check communications between PC and CPU board. If you cannot communicate, replace power supply.

Symptom	Possible Cause	Remedy
	Bad connection between internal terminal block and LED cable	Check connections
Low or no count from VFC on CPU board	PMT not connected properly	Check PMT connection
	No power to PMT	Check PMT cable connection from CPU board
	Incorrect or no control voltage applied to PMT	Check CPU board using PC in terminal mode
	Laser(s) not on or argon ion laser at too low a power setting	Run LIF light level diagnostics
	Optics module not aligned properly	Align optics module

Error Messages

When necessary, the BioFocus software performs self-diagnostics. If there are any pressure, temperature, electrical, or mechanical problems during a run, a message appears on the screen. Clicking through the first and any subsequent messages and taking actions suggested by these messages will usually eliminate the problem.

Error messages are also stored on the hard disk and can be viewed with a text editor like Windows Notepad. Error log files have the DOS extension LOG. Log files give a complete time stamped record of any errors that occur during the run. Table 7-3 is an alphabetical list of error messages and suggested course of action.

Table 7-3 Error Messages

Error Message	Action
Cannot recover from the Z180 NAK response (Firmware notification of a communications failure with the software.)	Reset the BioFocus by turning the power off and on, and then restarting the BioFocus software. Verify that both RS-232 cables are firmly connected at each end. If the problem continues, contact your Bio-Rad service department.
Carousel Compartment Door is Open	Verify the carousel door is closed securely.
Cartridge Compartment Door is Open	Verify cartridge door is closed securely.
Current is Out of Range	In constant current operation, the voltage limit setting may be too low to allow the power supply to source enough current. Increase the voltage limit setting.
Error: Inlet Lifter is DOWN	Reset carousels from the commands pull- down menu.

Error Message	Action
Error: Inlet Lifter is UP	Reset carousels from the commands pull- down menu.
Error: Outlet Lifter is DOWN	Reset carousels from the commands pull- down menu.
Error: Outlet Lifter is UP	Reset carousels from the commands pull- down menu.
High pressure is Above Range	The inlet gas pressure to your BioFocus is too high. Check the setting on your regulator.
High pressure is Below Range	The inlet gas pressure to your BioFocus is too low. Check the gas regulator setting and amount of gas in the tank. Check gas line for kinks or pinches.

Error Message	Action
Incoming LIC Fails to Match Computed LIC; ASCII Code (Decimal)	Reset the BioFocus by turning the power off and on, and then restarting the BioFocus software. Verify both RS-232 cables are firmly connected at each end. Make sure that the BioFocus instrument and the computer are connected to the same ground connection. If the problem continues, contact your Bio-Rad Service department.
Incoming Character is not an R (for response); ASCII Code (Decimal)	
Incoming Command Character Failure; ASCII Code (Decimal)	
Incoming LIC High Digit Failure; ASCII Code (Decimal)	
Incoming LIC Low Digit Failure; ASCII Code (Decimal)	
Incoming Sequence Character Failure; ASCII Code	
Incoming STX Character Failure; ASCII Code (Decimal)	
Inlet Carousel Problem	Reset carousels and run carousel diagnostics (Commands).
Inlet Lifter Problem	Reset carousels on turntables and run carousel diagnostics.
Insufficient Cartridge Coolant Flow	Check the coolant fluid level and make sure there are no air bubbles coming from the reservoir filter. Replace reservoir filter; check that hole filter end is up.

Error Message	Action
Low pressure is Below Range	The inlet gas pressure to your BioFocus is too low. Check the gas regulator setting and amount of gas in the tank.
NAK to the Z180—Software notification of a communications failure	Reset the BioFocus by turning the power off and on, and then restarting the BioFocus software. Verify both RS-232 cables are firmly connected at each end. If the problem continues, contact your Bio-Rad Service department.
Outlet Carousel Problem	Reset carousels and run carousel diagnostics.
Outlet Lifter problem	Reset carousels and run carousel diagnostics.
Timeout waiting for the Z180 ACK/NAK Response (Firmware not responding to software.)	Reset the BioFocus by turning the power off and on, and then restarting the BioFocus software. Verify both RS-232 cables are firmly connected at each end. If the problem continues, contact your Bio-Rad Service department.
Timeout waiting for the Z180 BEL/SYN response. (Firmware is not responding to a software command.)	
Undefined Response Code; ASCII Code (Decimal)	
Voltage is Out of Range	In constant current operation, the voltage limit setting may be too low to allow the power supply to source enough current. Increase the voltage limit setting.

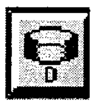
Error Message	Action
Z180 response is a syn at the EOT level. (A purge, injection or focusing command has failed to complete properly.) Z180 response is not the expected ACK or NAK; ASCII Code (Decimal): (Firmware response to a communication is incorrect.) Z180 response is not the expected BEL or SYN; ASCII Code (Decimal): (Firmware response to a command is incorrect.) Z180 response is not the expected EOT or SYN; ASCII Code (Decimal)	Reset the BioFocus by turning the power off and on, and then restarting the BioFocus software. Verify both RS-232 cables are firmly connected at each end. If the problem continues, contact your Bio-Rad service department.

Diagnostics Screens

BioFocus software has a complete set of diagnostic operations you can use to verify proper system operation. These diagnostics can be accessed by clicking the appropriate button from the main BioFocus software screen. Alternatively, under the Commands Menu, there is a Diagnostics section. Clicking this will produce a box listing Carousels, Light Levels, Pressure, Electrical, and Temperature.

If you click one of these screens, you can test various instrument functions.

Carousel Diagnostics



Carousel Diagnostics Button

If you suspect that a vial holder or carousel is not working properly, you may wish to check the carousel function outside of an automation sequence. Refer to the screen shown in Figure 7-1.

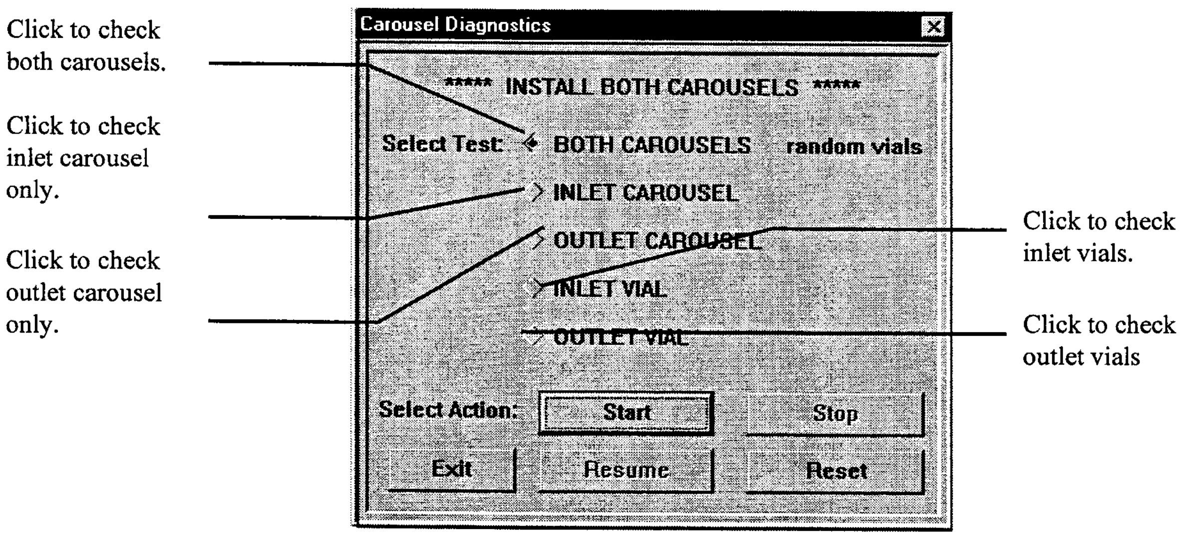


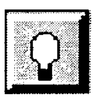
Figure 7-1 Carousel Diagnostics Screen

There are five carousel tests and 10 control buttons (described below). Click the button to start (or stop) a test.

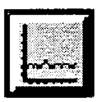
- **BOTH CAROUSELS** (default) – each position on each carousel is tested once during a test cycle of 32 per carousel. The test alternates regularly between the inlet and the outlet carousel and positions are tested in pseudo-random order.

- **INLET CAROUSEL** – each position is tested in order on the inlet carousel.
- **OUTLET CAROUSEL** – each position is tested in order on the outlet carousel.
- **INLET VIAL** – the selected vial position on the inlet carousel is tested.
- **OUTLET VIAL** – tests the selected vial position on the outlet carousel.
- **START** (default) – start continuous operation of the selected test.
- **STOP** – stop continuous operation of the test that is in progress.
- **RESUME** – resume a user-stopped test at the point where it stopped; this will not resume an error-stopped test.
- **RESET** – reset the vial holders to their lower positions and the carousels to position 1 (home).
- **EXIT** – end carousel tests.

Light Level Diagnostics (Absorbance Mode)



Light Level Diagnostics Button



Monitor Baseline Button

Since baseline noise doubles for a fourfold reduction in light, noise may be indicative of a deteriorating lamp. The detector provides the capability of monitoring relative light intensities at both the sample and reference photodiodes. If an unusually noisy baseline is noted under Detector Baseline Monitor while high voltage is off, intensities of reference and sample light should be assessed. The detector baseline monitor is accessed from the Commands menu or a button on the main menu bar.

To measure total noise, use the test cartridge supplied with the BioFocus. Install the test cartridge and perform a single wavelength run with zero volts. (Because the test cartridge has no capillary, there is no need for any buffers.) A 60 minute run can be used to determine noise and drift.

Light level diagnostics serve to measure light intensities from either the D_2 lamp or the tungsten lamp. Access the light level diagnostics by clicking the light level diagnostics button at the top of the main screen. See Figure 7-2.

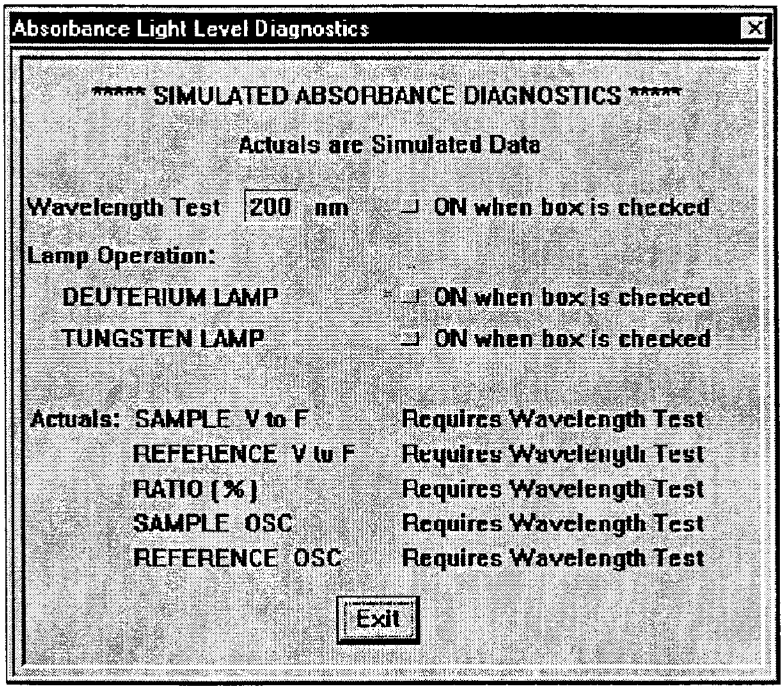


Figure 7-2 Absorbance Light Level Diagnostics Screen

To check light levels, enter a wavelength in the box beside Wavelength Test and click the "**ON when box is checked**" on the same line. This will turn on the appropriate lamp: 190 nm through 365 nm turns on the deuterium lamp; 366 nm through 800 nm turns on the tungsten lamp, if one is present. If you are testing a wavelength in the 366-800 nm range, be sure that the deuterium lamp is off (it may have been left on from an earlier test).

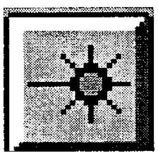
Under SAMPLE V to F you will see a number displayed which represents the sample light level. A Sample V to F minimum reading of 8 means no light for a BioFocus 3000. A reading of 16 means no light for a BioFocus 2000. A typical value is greater than 30. This number changes, depending on the cartridge and wavelength. Testing without a cartridge will give a high light level, generally close to the reference reading.

You do not need to be concerned with the last two OSC values, the ratio of the V to F values on this Diagnostics screen, but they may be requested during a telephone conversation with a Bio-Rad Service representative.

As lamps age they show a gradual decrease in light levels across the entire spectrum. In multiwavelength operation light levels on the reference must be above 40 or a low light level error will occur. In high speed scanning, the reference must be above 8 across the entire spectrum or a low light level error will occur. When running baseline monitor in a BioFocus 3000 system, the default setting uses high speed scanning. This means that even if you are looking at 250 nm, the instrument is collecting data across the entire UV spectrum.

To quit the Light Level Diagnostics screen, **turn** off the Wavelength test (click the box to remove the check mark and turn the lamp off). Then click **EXIT**.

Light Level Diagnostics (LIF Mode)



LIF Light Level Diagnostics Button

System operating software includes diagnostics for the fluorescence detector. Figure 7-3 shows the diagnostics screen that appears when you click the **LIF Light Level Diagnostics** button.

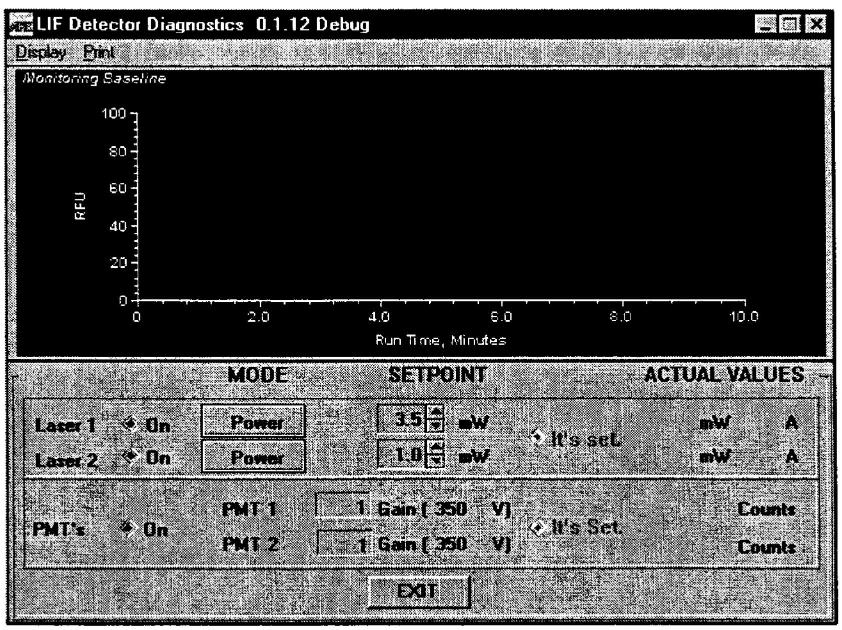
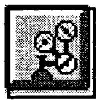


Figure 7-3 Fluorescence Detector Diagnostics Screen

From this screen you can control the laser(s) and photomultipliers. You can read photomultiplier outputs in A/D counts on a scale of 0 to approximately 200,000.

Pressure Diagnostics



Pressure Diagnostics Button

If you have any concern that purge or sample injection pressure is below or above the programmed value for one or more vial positions, you may test these functions in the Pressure Diagnostics screen shown in Figure 7-4.

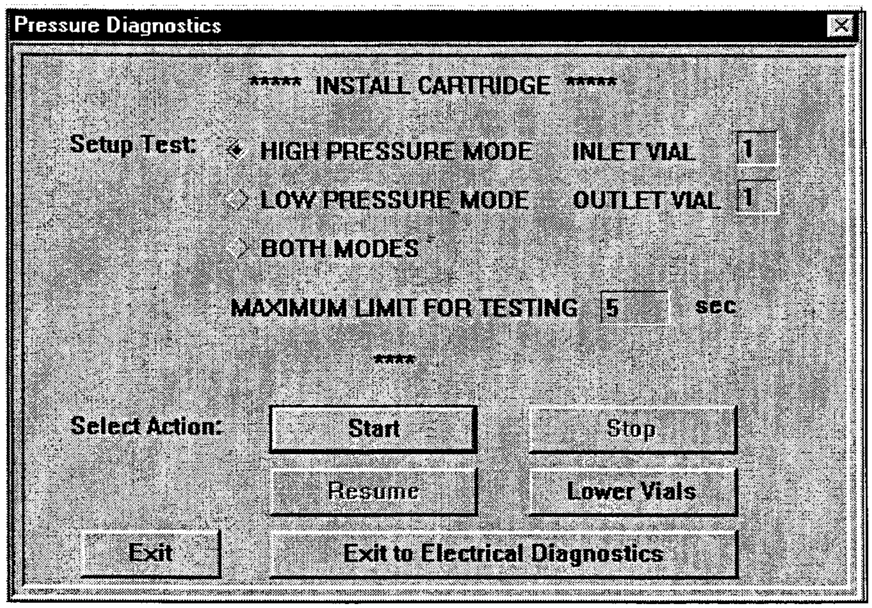


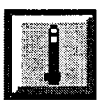
Figure 7-4 Pressure Diagnostics Screen

There are three pressure tests and five control buttons, described below.

- **HIGH PRESSURE MODE** (default) – a timed high pressure test (stops automatically), with a maximum time of 5999 seconds (99.98 minutes). Enter a maximum limit for testing in the seconds box. You must specify which inlet and outlet vial positions are to be tested. In addition to diagnosing a problem, this option allows you to purge a capillary without running an automation sequence.
- **LOW PRESSURE MODE** – a controlled low pressure test (stops automatically), maximum value is 500 psi*seconds. Enter a maximum limit for testing in the psi*seconds box. You must specify which inlet and outlet vial positions are to be tested. During a low pressure test, the actual pressure is displayed and automatically updated.

- **BOTH MODES** – a controlled low pressure test followed by timed high pressure test; test repeats until user clicks **STOP**. There are two choices: (1) test each inlet position sequentially (default), or (2) test only one selected inlet position; maximum limits are 99 seconds for high pressure or 498 psi*sec for low pressure.
- **START** (default) – start a controlled pressure test—lowers lifters if they are up.
- **STOP** – stop the test in progress; lifters remain up.
- **RESUME** – repeat a test. Control values start at zero.
- **LOWER VIALS** – lower the lifters.
- **EXIT** – quit the Pressure Diagnostics screen.
- **EXIT to Electrical Diagnostics** – stops pressure test and brings up the electrical diagnostic screen.

Temperature Diagnostics



Temperature Diagnostics Button

With the BioFocus TC Models, the Temperature Diagnostics screen displays three temperature regions (cartridge, carousel and ambient) and five control buttons. See Figure 7-5.

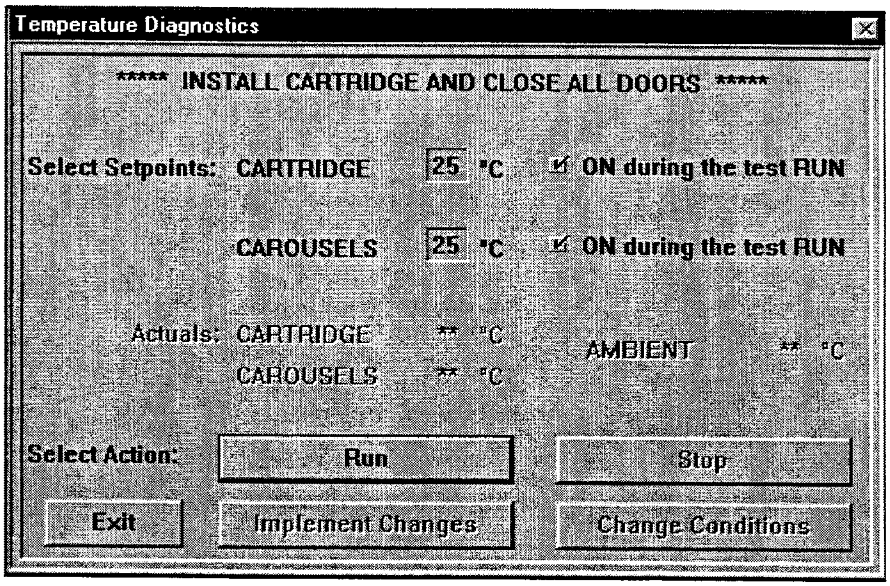


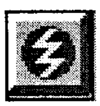
Figure 7-5 Temperature Diagnostics Screen

- **CARTRIDGE** region tests 15°C through 40°C (same options for programming an automation sequence)

- **CAROUSELS** region tests 4°C through 40°C (same options for programming an automation sequence) This is only available on units with carousel temperature control.
- **AMBIENT** region; monitored only (not user programmable). Measures waste heat sink temperature.
- **ON during the test RUN** when checked, controls the associated region to the specified temperature; when not checked, the associated region is not temperature controlled. These commands are implemented immediately during a RUN.
- **RUN** (default): run a continuous test of all temperature regions.
- **STOP**: stop all temperature tests in progress
- **CHANGE CONDITIONS**: enter a different temperature test value in any of the three boxes above (this does not stop the other tests in progress).
- **IMPLEMENT CHANGES:** run the changed conditions (new temperature value)
- **EXIT:** quit the Temperature Diagnostics screen.

During testing, the actual temperatures are displayed and updated automatically. When the test undergoes a user-requested exit, control for the cartridge stops. In BioFocus TC models the carousels region is reset to its user setpoint temperature, and control for the cartridge stops.

Electrical Diagnostics



Electrical Diagnostics Button

If you suspect a voltage or current problem for a single run based on the appearance of the electropherogram, you may wish to review the current and voltage traces for the length of the run, looking for any abnormal fluctuations. Comparing traces with other runs in an automation sequence may reveal that the problem was an isolated event due to poor capillary purging or a contaminated reagent or sample.

If symptoms only occur when working with a particular User-Assembled Cartridge, verify that there is no coolant leakage and replace the inlet silicone seal before testing. If the same symptoms recur after replacing the cartridge and reagents, you should perform Electrical Diagnostics using either the high voltage test cartridge (provided in the BioFocus accessories kit) or the capillary cartridge and run buffers of interest. If you use the latter, be sure to have the run buffer vials in position prior to running Electrical Diagnostics.

When symptoms of electrical problems occur under various run conditions and with different cartridges, there may be electrical arcing outside the cartridge itself. Sharp baseline spikes are usually observed when arcing. Other symptoms of arcing are "Current Leakage" messages and in extreme cases, popping, frying, or hissing sounds at higher voltages.

In many cases the high voltage test cartridge should be used to rule out buffers and damaged capillaries as the source of high voltage or current fluctuations. The test cartridge also allows easy testing of baseline noise, because it contains no lens, only a clear detector window. If the system is working properly, there should be a close to zero current reading since the circuit is not complete through the high voltage test cartridge. (Note that the coolant path through this cartridge is functional.)

Cleaning the Electrode Area

NOTE: Buffer contamination is the primary cause of high voltage arcing. To clean the electrode area, remove the three thumb screws that hold the cover in front of the capillary in place and remove the cover. This exposes the inlet electrode area from top as shown below. Clean the area using water and a cotton swab.

Before beginning any high voltage troubleshooting, clean and dry the inlet pressure chamber and electrode and the bottom of the cartridge holder to eliminate high voltage arcing from any contaminants. See Figure 7-6.

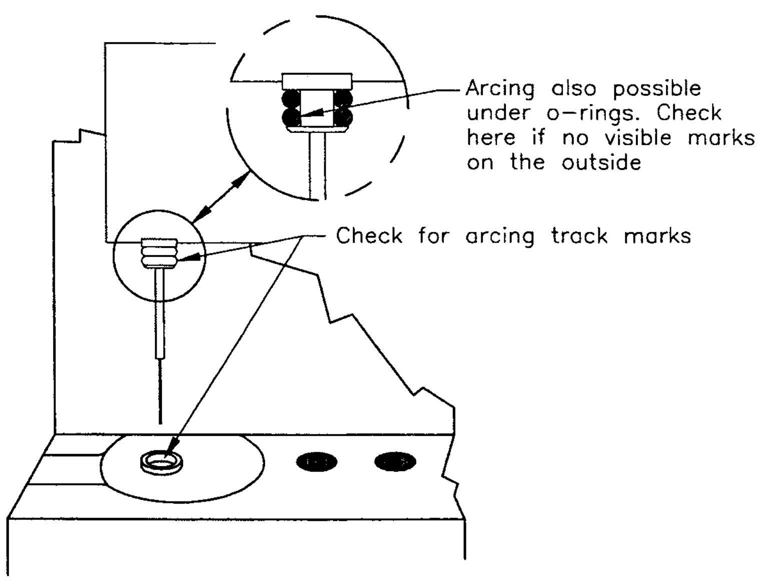


Figure 7-6 Inlet Electrode Area

Failure to keep this area clean can result in electrical current leakage errors and in some cases, arcing.

Inserting High Voltage Test Cartridge

Insert the high voltage test cartridge or the cartridge of interest into the cartridge holder. With the high voltage test cartridge no buffers are required. Be sure that the O-rings on the test cartridge are in good condition. Avoid touching the O-rings on the test cartridge with bare hands because skin oils and salts may cause arcing.

CAUTION! Never run Electrical Diagnostics without a cartridge inserted. Arcing may damage the inlet pressure chamber and electrode.

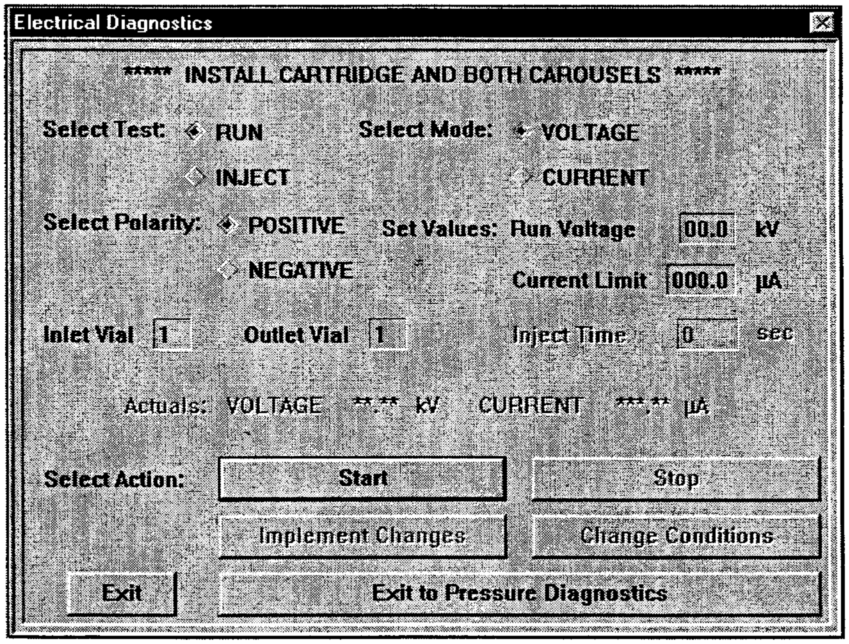


Figure 7-7 Electrical Diagnostics Screen

Running Diagnostics

Refer to Figure 7-7. Select **VOLTAGE** or **CURRENT** and select Polarity. Select **RUN** unless you wish to test electrophoretic sample injection (if you select **INJECT** you must also enter an **INJECT TIME** from 1-5999 seconds). During a test the actual voltage or current will be displayed. If you are testing constant voltage, enter a voltage test value (kV) and a current limit value (µA)in the current box. If you are testing constant current, enter a current test value (µA) and a voltage limit value (kV) in the voltage box. Click **START**.

NOTE: A monitor baseline test may be performed while running electrical diagnostics. This is a useful indicator, because leakage or arcing creates noise on the baseline.

With the test cartridge, use 30 kV constant voltage and 50 µA current limit as test values. After the test begins, you should observe a reading of 30 kV ±1 kV. The current reading should be less than 0.5 µA and it should not fluctuate more than 0.02 µA. If you do not obtain these results, you may have high voltage arcing around the inlet pressure chamber and electrode.

Detector Autocalibrate (BioFocus 3000 Only)

When the detector needs to be autocalibrated, the main program posts a screen message to the user. Under normal conditions, detector autocalibration is not required unless requested by a Bio-Rad service representative.

The user should initiate a detector autocalibration from Diagnostics under the Commands menu if both of the following conditions are satisfied:

1. The detector has been on continuously for at least one hour and,

2. The detector is currently idle, i.e., not collecting data, not monitoring the baseline, and not undergoing Light Level Diagnostics.

Electrical Current Problems

At times you may get low or no current through the capillary. During a run, if the voltage shown is correct, the most probable cause of low current is the buffer or the capillary. Make sure there is enough buffer in the vials and an adequate purge time is used.

The quickest test is to purge the capillary for several minutes with buffer, then run the electrical diagnostics to check the current. (Purging can be done with the pressure diagnostics screen described earlier in this chapter.) If the current increases, but is still low, purge the capillary longer.

Air Bubbles in Capillary or Vial

Low or no current can be caused by an air bubble in the capillary. When pipetting into a vial, be sure an air pocket does not get caught in the bottom of the vial. The capillary end will be in the air pocket and no current will flow. To degas buffers, spin them for at least three minutes in a centrifuge. Excessive current in the capillary can cause Joule heating and create a gas bubble also, so be careful when running high current separations.

Blocked Capillary

If the current is still low you may have a blockage in the capillary. Often the ends of the capillary dry out and any residual salt can crystallize, causing a blockage. Capillaries can often be unblocked by soaking the capillary ends in water, or by sonication.

Alternatively, you might clear the blockage with the capillary purge device. The purge device provided is a 1 ml syringe with an adapter that can be inserted over the capillary inlet. See Figure 7-8. Fill the syringe with deionized water and insert it into the clear plastic luer fitting on the end of the adapter. Push the two flanges on the back of the capillary cartridge in, and carefully push the top of the cartridge downward, exposing both ends of the capillary. Carefully insert the adapter over the capillary inlet (covered by PEEK) and press the adapter firmly against the cartridge O-rings. Snap the metal retainer clip over the top of the cartridge for a tight fit. Push the syringe plunger down gently and slowly—do not use excessive force. Observe whether a drop of fluid exits the capillary on the outlet side. The drop will appear after 10-20 seconds and may be barely visible. If no drop is seen, you may try to clear the blockage by soaking the capillary ends in hot water. If that does not open the capillary, then it must be replaced.

CAUTION! Rough handling of the exposed capillary ends or excessive purge pressure may result in capillary breakage.

NOTE: Deionized water does not conduct electricity. After purging with deionized water, you must purge again with running buffer before running electrical diagnostics.

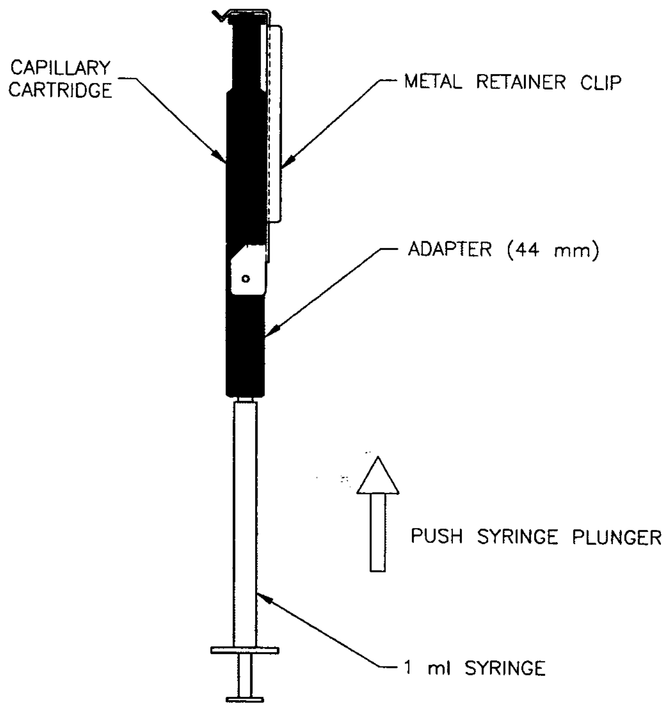


Figure 7-8 Capillary Purge Device Operation

Damaged Electrodes

Damaged electrodes will also cause a low (or no) current condition. Inspect the inlet and outlet electrodes. Electrodes may be viewed from the carousel compartment by use of a mirror and flashlight directed at the compartment ceiling, or by looking down from the cartridge compartment.

Make sure the electrodes are aligned to the center of the vial, and are straight. If vials are not fully inserted into the vial holder, they can damage the electrodes. In the event that an electrode is broken, there may be no current or intermittent current during a run.

A cross section of the pressure chamber and inlet electrode (shown in Figure 7-9) shows their positions relative to an elevated carousel vial holder and the capillary inlet.

NOTE: At the lower rim of the pressure chamber there are two seals that will wear over time, necessitating eventual pressure chamber replacement for efficient purging and sample pressure injection. These seals should be cleaned with a cotton swab and water every month to ensure proper operation.

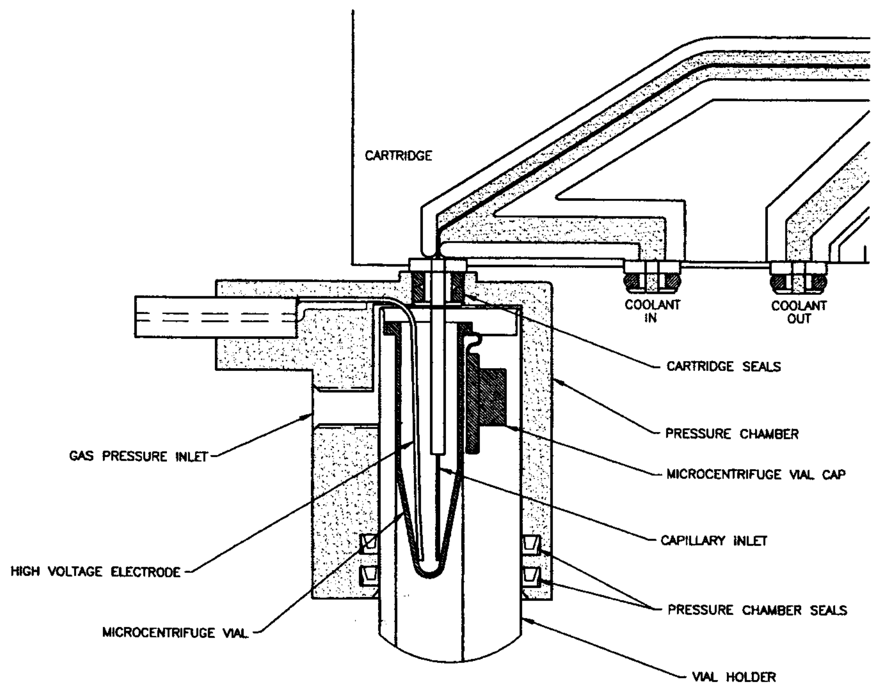


Figure 7-9 Inlet Electrode/Pressure Chamber Assembly Cross Section

The following is a list of Bio-Rad catalog numbers for inlet and outlet replacement electrodes and kits:

Pressure chamber with inlet electrode:	148-6057
Outlet electrode	148-6058
BioFocus inlet electrode pack	148-6085

To replace the pressure chamber and inlet electrode, turn off the power switch on the rear panel of the BioFocus system. Open both doors and remove the inlet carousel. Remove the capillary cartridge from the cartridge holder. Unscrew the black thumb screws on the front of the cartridge holder and remove the front plate. You can now see the top of the pressure chamber to the left. Pull out the two black plastic pins near the lower left corner of the cartridge holder to release the pressure chamber. Remove the lamp cover. Then unscrew the gas connection and unplug the electrode connection on the right side of the cartridge holder. Reaching up through the carousel compartment, pull down the entire pressure chamber with its inlet electrode. Pull the high voltage connector and gas line down through the cartridge holder.

Feed the high voltage connector and pneumatic line of the new pressure chamber up through the cartridge holder. Insert the pressure chamber back through the ceiling of the carousel compartment and into the base of the cartridge holder. Be careful not to pull on or bend the platinum electrode inside it. Reconnect the high voltage electrode and gas line to their appropriate connections on the right side of the cartridge compartment. Push the two black plastic support pins back into the cartridge holder assembly. Replace the lamp cover.

To replace the outlet electrode, turn off the power switch on the rear panel of the BioFocus. Open both doors and remove the outlet carousel. Remove the capillary cartridge from the cartridge holder. Unscrew the black thumb screws on the front of the cartridge holder and remove the front plate. Loosen the silver knurled nut on the lower right side of the cartridge holder two or three turns. This releases the electrode.

Using a hemostat, tweezers, or a small pair of needle nose pliers, reach through the ceiling of the carousel compartment, grasp the outlet electrode and pull to the left until the electrode is free.

Once the electrode has been removed, take the replacement electrode and insert the long end into the outlet electrode changing tool. Place the upper end of the electrode in the small groove at the top of the tool (see Figure 7-10). Insert the tool with the electrode into the outlet opening in the carousel compartment. Looking down though the cartridge holder, rotate this tool until the groove points to the right, where there is a small hole for the top end of the outlet electrode. The hole is in the upper corner of the chamber. Once it is properly aligned, push the tool to the right, inserting the top end of the electrode into the hole. It may be necessary to move the knurled nut in and out while inserting the electrode. Tighten the silver knurled nut on the lower right side of the cartridge holder to secure the electrode.

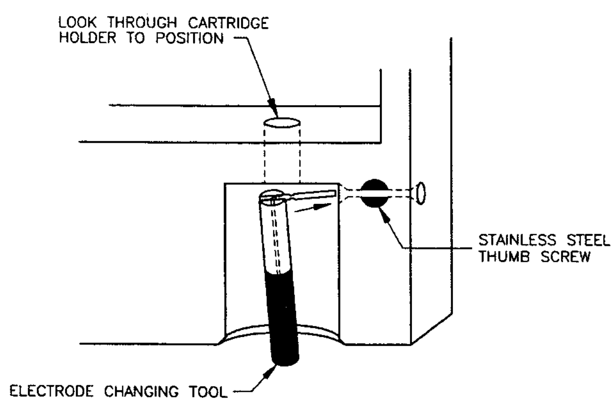


Figure 7-10 Using the Outlet Electrode Changing Tool

Replaceable Inlet Electrodes

Newer model BioFocus systems have inlet electrodes that can be replaced. To replace the electrode, remove the pressure assembly as described previously. Then follow the instructions below.

Before installing an inlet electrode, remove the old one by gripping it with tweezers or small pliers and pulling it straight out the bottom of the pressure chamber area.

Installing a new inlet electrode is a three-step process:

1. Push the inlet electrode all the way into the electrode insertion tool.

2. Push the inlet electrode all the way into the socket inside the pressure chamber

3. Use the inlet electrode insertion guide to position the electrode. Remove the inlet electrode insertion tool before removing the inlet electrode insertion guide.

High Voltage Problems

One possible high voltage problem is arcing. Arcing can manifest itself in a variety of ways. One common sign is a series of small spikes on the electropherogram. Severe arcing can cause electrical problems that appear to be computer or software problems. Often arc damage can be seen along the top of the pressure chamber or on the orange O-rings at the high voltage end of the capillary.

To prevent arcing, be sure the cartridge holder assembly is very clean. Clean up any spilled buffer immediately--especially around the high voltage electrode. Cooling water that has become contaminated can also cause arcing. Use only deionized water or Fluorinert. Put 20% isopropanol or methanol in the cooling water to prevent bacterial growth.

Areas damaged by arcing can often be cleaned to remove carbonized contamination. If cleaning these areas does not solve the arcing problem, the affected parts must be replaced. Figure 7-11 shows where to look for damage.

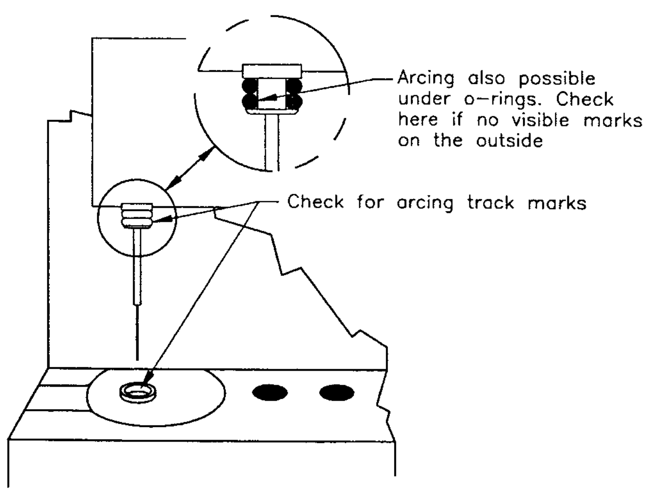


Figure 7-11 Areas to Look for Arc Damage

Coolant System Problems

NOTE: Do not use vacuum grease on coolant O-rings or liquid couplings. The grease will be pulled throughout the system and will clog components.

To properly maintain the cartridge cooling system on the BioFocus always use deionized water or Fluorinert. To eliminate bacterial growth when using water as a coolant, add approximately 20% isopropyl alcohol or methanol to the coolant liquid reservoir.

Air Bubbles in Coolant Reservoir

Check the coolant reservoir and be sure there is no more than occasional air bubbles coming through the coolant. The coolant system will work well even with a small amount of air bubbles in the coolant system. If you are not getting a cooling error during the run, the coolant system is working adequately. Excess air bubbles can dramatically increase the evaporation rate of the coolant.

If air bubbles exist, first re-seat the cartridge. If you are using a User-Assembled cartridge, change to the test cartridge and see if the bubbles are reduced. If the test cartridge reduces bubbling, there may be an air leak in the User-Assembled cartridge. Refer to the User-Assembled cartridge instructions and review the assembly process.

A shunt tube can be substituted for the cartridge to differentiate coolant system problems from cartridge-cartridge holder interface problems.

If the air bubbles continue, check the seals on the cartridge and make sure they are not damaged. If the seals are good, you may need two new liquid couplings (part # 920-5855). Liquid couplings are easy to replace. They are located on the cartridge holder assembly where the coolant flows into the cartridge. To replace them, loosen the black plastic pins from the front of the cartridge holder by pulling the pins forward. Pull out the liquid couplings by reaching into the carousel area and pulling down. (Do not cross the couplings—keep the coolant path through the cartridge with the inlet on the left, and the outlet on the right.)

When you replace the couplings, it is a good idea to trim back some of the tubing to create a better seal around the barb on the coupling. Trim off about 5 mm of tubing before sticking the new coupling on to the tubing. Reinsert the coupling and push in the black plastic pins.

Plugged Capillary Coolant Path

Your system includes a purge device that can be used to check for possible coolant path plugging in the capillary cartridge. This device has a short adapter attached to the end of a 20 ml syringe and another adapter with 12 in. (30 cm) of polyurethane tubing attached. You will need a small beaker of coolant (deionized water or Fluorinert) to perform a test. See Figure 7-12.

To check the coolant path on a user-assembled cartridge, insert the syringe adapter over the inlet coolant port. Insert the tubing adapter over the outlet coolant port and place the open end of the tubing into the beaker of coolant. Withdraw the plunger of the syringe and observe whether any coolant enters the syringe.

CAUTION! Never push the plunger inward—high pressure may cause the sealed areas to leak.

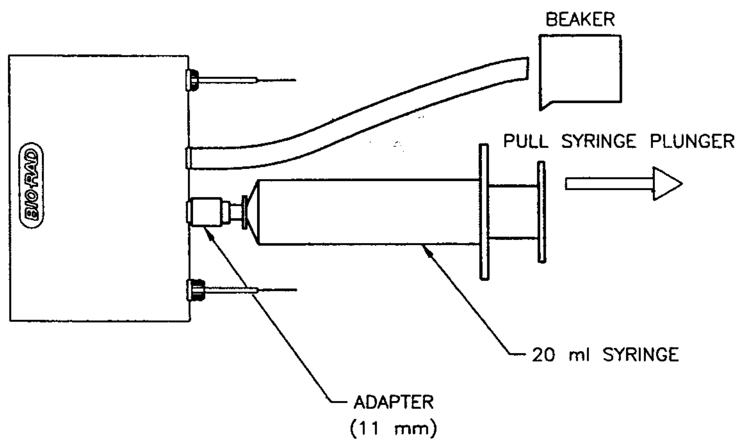


Figure 7-12 Operation of Cartridge Coolant Purge Device

Noisy Baselines and Sensitivity Problems, Absorbance

Noisy baselines can be caused by a variety of problems. The most common source of noise is an old deuterium lamp. High voltage arcing can also cause baseline noise. The best method for finding the cause of a noisy baseline is to use the test cartridge provided with your instrument.

The test cartridge should be used when the voltage or current readings on the BioFocus are unstable. The test cartridge eliminates buffers and cartridges as the source of high voltage or current fluctuations, and allows easy testing of the high voltage electronics. The test cartridge also allows easy testing of baseline noise, because it eliminates the optical flow cell and replaces it with a clear window. Before any high voltage troubleshooting can begin, clean the inlet electrode assembly to eliminate high voltage arcing.

CAUTION! Do not run electrical diagnostics without a cartridge or carousels inserted, because high voltage arcing may damage the high voltage electrode assembly.

Place the test cartridge into the cartridge holder assembly as you would a regular cartridge. Because there is no capillary, no special buffers or configurations are required. Be sure the O-rings on the test cartridge are in good condition. Do not touch the O-rings on the test cartridge with bare hands, because oils and salts may cause arcing.

Current Leakage Test

First, it is important to determine the source of the noisy baseline.

- Turn on the instrument. It may be necessary to let the instrument warm up for half an hour before making noise measurements to reduce detector baseline drift.
- Make sure that the inlet and outlet carousels are in the instrument.
- Click the **SRG** button on the main menu or select Single Run Group from the Commands menu. Then select CZE method.
- Change the following parameters:

Run Parameters

- Injection pressure*time – 0
- Run Voltage – 0
- Purge to – 0 purges

- Run time – 2.00 minute(s)

Detector Parameters

- AU Range to 0.0005 (if recorder is being used)
- Display
- AU range to 0.0005

- Click **Start**
- Increase the screen to full.
- Measure the noise from peak to peak with a scale. Record the maximum measurement.
- It should be less than 20% of full scale or 100 μAU.
- If it is not, then the problem is with the detector or lamp.
- If the baseline is acceptable, then select Single Run Group again.
- Change the Run Voltage to 25 kV and click **Start**.
- Record the maximum baseline noise as before. It should be very close to the measurement previously taken.
- If it is not, then there is an electrical leakage or arcing problem. The problem can usually be corrected by cleaning or replacing the Pressure Chamber/Inlet Electrode. Other sources could be dirty vial holders, dirty high voltage connectors or a test cartridge that has developed a conductive path.
- A noisy baseline can also be caused by loose pieces on the ground potential electrode or by cartridge problems. These problems would not show excessive noise on the previous tests.

Current, Voltage, and Absorbance Stability Test

Once the run is finished, convert the data using the integrator conversion program. Plotting the high voltage and current will show any instabilities in the high voltage circuits. Scaling is direct 1% per kV and 1% per μA; 30 kV will be 30% of full scale, and 10 μA will be 10% of full scale.

NOTE: A 45-minute warm-up is required to test noise and drift accurately.

Plotting the absorbance will show system noise and drift. If the absorbance noise or drift exceed the system specifications, a new deuterium lamp may be required.

Carousel Vials and Vial Cap Replacement

Inserting a carousel into the carousel compartment without making sure that all vial holder caps are closed may result in cap breakage. Spare vials and vial holder cap assemblies have been provided with your system and are easy to replace.

Using the small hex wrench provided, remove the two screws on the carousel cover and lift the cover up (it may be necessary to use the end of the wrench as a lever).

NOTE: After you remove the carousel cover, keep the carousel on a solid surface to avoid having vials fall out the bottom.

Remove the broken cap assembly and the vial holder will drop down. Push the vial holder up approximately 2 centimeters above the carousel surface. With your other hand, insert a new vial cap assembly where the old vial cap assembly was, and let the vial holder drop down. It will catch the vial cap hinge and the vial cap should drop down on top of it. Replace the two screws holding the carousel cover. Make sure that the arrow points to Position 22 for the Inlet Carousel or Position 12 for the outlet carousel.

Appendices

Warranty

The BioFocus system is warranted for a period of one year against defects in materials and workmanship. If any defects should occur in the system during this period, Bio-Rad Laboratories will repair or replace the defective parts free of charge. For the exact terms of warranty, please see the Instrument Warranty Card.

Defects caused by the following actions are specifically excluded:

1. Improper system operation or system abuse.
2. Repair or modification of the system performed by anyone other than Bio-Rad Laboratories or its authorized agent.
3. Use of fittings or other spare parts not authorized by Bio-Rad Laboratories.
4. Inappropriate interfacing to external devices.
5. Use of inappropriate solvent or sample. (Most organic solvents can be used at concentrations up to 70% in aqueous buffers.)
6. Non-system related facility problems.

This one year warranty does not apply to parts listed below:

1. Detector lamps
2. Capillary cartridges
3. Electrodes
4. Fuses
5. Computer purchased outside of Bio-Rad Laboratories

For inquiries or requests regarding system repair or service, contact your local Bio-Rad office or distributor (in the US call Technical Service 1-800-424-6723). Please have the serial number available. It is located on the rear panel of your BioFocus system. The software version numbers may also be requested, so please check the "About ..." (located under Commands in the BioFocus software and under File in the Integrator software) and have this information available with your serial number.

Glossary of Terms and Abbreviations

The definitions in this glossary pertain only to their usage in this manual.

Ampholytes	A multicomponent mixture of amphoteric compounds that can form a pH gradient in an electric field.
AUFS	Absorbance Units Full Scale.
Buffer	Aqueous solution of salts and buffering agents used in electrophoresis.
Capillary	Fused silica (quartz) tubing of special design for capillary electrophoresis.
Capillary Cartridge	The holder for the capillary in the form of an insertable cartridge, includes coolant ports and prealigned optics for the detector.
CE	Capillary Electrophoresis.
CZE	Capillary Zone Electrophoresis. Also known as free zone electrophoresis. Electrophoretic separations performed in free solution without the presence of a support structure such as a gel.
D_2 Lamp	Deuterium lamp.
Dynamic Sieving	A capillary electrophoresis method that incorporates linear polymers into the buffer solutions so that analytes can be separated by size.
EEO	Electroendosmosis, also known as electroosmotic flow, or "EOF". Bulk fluid flow through the capillary during application of high voltage caused by the negatively charged surface of uncoated fused silica. EEO flow is usually towards the cathode and occurs above pH 3. When the capillary is treated chemically to create a positively charged surface, EEO flow will be towards the anode.
Electrode	On the BioFocus systems, platinum wire used to introduce high voltage into the buffer solutions in the buffer vials.

Electroendosmotic Flow	See EEO.
Electrolyte	Aqueous solution that is electrically conductive.
Electropherogram	The trace obtained by running capillary electrophoresis.
Electrophoretic Mobility	The velocity with which a charged particle moves in an electric field of a given field strength. Electrophoretic mobility is determined by a particle's mass and the net charge that it carries.
Error Messages	Messages that appear on the computer screen when a condition exists that prevents normal operation of the BioFocus system.
Field Strength	Electric potential over distance. Expressed as volts per centimeter of capillary.
Focused	When a sample has reached its point of electrical neutrality in isoelectric focusing.
Free Zone	Indicates free solution electrophoresis as opposed to systems with gels or other supports. Also known as Capillary Zone Electrophoresis (CZE).
HPCE	High Performance Capillary Electrophoresis. Applies to all forms of capillary electrophoresis.
ID	Inside Diameter (usually given in micrometers) of the capillary. Can also refer to an Identifying name in the BioFocus software.
IEF	IsoElectric Focusing. Isoelectric focusing is a method by which molecular species are separated by differences in their isoelectric points. In the BioFocus system, IEF refers to the use of ampholytes to create a pH gradient within the capillary. Application of an electrical potential causes the analytes to migrate or "focus" in the region of the pH gradient in the capillary where they appear to be neutral.

Inject	The term used to describe the application of high voltage or pressure for a short time to cause migration of sample into the capillary.
Inlet Vial	The microcentrifuge tube that holds buffer, wash or sample at the inlet end of the capillary.
Inlet	The end of the capillary where samples are introduced and purges begin.
Inlet Electrode	The high voltage electrode located on the inlet side of the cartridge compartment.
kV	Kilovolts.
LIF	Laser-induced fluorescence.
MEKC	Micellar Electrokinetic Capillary Chromatography. A capillary electrophoresis method combining elements of both electrophoresis and chromatography.
Micelle	An aggregate of surfactant molecules in solution.
Mode	The functional condition of the high voltage power supply. Can be either constant voltage or constant current.
mV	Millivolts
nm	Nanometers
NPT	Pipe thread couplings for gas tank connection. Sometimes preceded by F (for female) or M (for male).
OD	Outside diameter (usually in micrometers) of capillary.
Outlet	The end of the capillary where analytes migrate out and purge reagents empty into the waste vial.
Path length	The distance through the sample solution at the point of detection (capillary ID).
Photodiode	Solid state light sensing device.
PMT	Photomultiplier tube, an electronic light sensing device.
Purge	Forcing fluid through the capillary.

Rise Time	The time it takes for the recorder signal to rise from a 10% response to a 90% response. It is a function of an electronic filtering method.
Run	Performance of a capillary electrophoresis analysis or separation.
RFU	Relative Fluorescence Units, defined as the fluorescent signal produced by a 50 picomolar fluorescein concentration in a 75 µm capillary.
UV/Vis	Ultraviolet/Visible light.
V/AU	Volts per Absorbance Unit.
Vial Holder	Plastic cylinder that holds the disposable microcentrifuge tube in a BioFocus carousel.
Zone	Distance or volume occupied by the sample in the capillary during a CZE separation.
µA	Microamperes.
µm	Micrometers.